

Chenyang Zhang

cyzhange@stu.ecnu.edu.cn | IM-DM4AI

 in/ chenyang-zhang-109372313 |  lovelynewlife |  0009-0007-8265-8461

No. 3663 Zhongshan North Road, Shanghai - 200062, China

BRIEF INTRODUCTION

I am a Ph.D. candidate in Software Engineering at the School of Data Science and Engineering, East China Normal University. My research focuses on systems for AI and AI-driven system optimization, with an emphasis on execution efficiency, data movement, resource scheduling, and system-level automation. I currently study efficient AI execution in data systems, including database-native model inference, recommendation workloads, and agent-driven high-performance database optimization. Looking forward, I am interested in using AI to automate system research workflows, including workload analysis, optimization search, implementation, and performance profiling.

EDUCATION

- **East China Normal University** September 2022 - Present
Ph.D. Candidate in Software Engineering, Data Management and Intelligence Computing (DMIC) System Group Shanghai, China
 - GPA: 3.52/4.00, ranked in the top 5%.
 - Conducting research on systems for AI and AI-driven system optimization, advised by Prof. [Chen Xu](#).
- **Donghua University** September 2018 - June 2022
B.Eng. in Software Engineering Shanghai, China
 - GPA: 4.13/5.00, ranked in the top 5%.
 - Outstanding Undergraduate Thesis Award.

EXPERIENCE

- **Microsoft Research Asia (MSRA)**  May 2026 - Present
Research Intern Beijing, China
 - Exploring agent-driven database engine specialization, using workload understanding and query code synthesis to generate optimized execution paths and reduce general-purpose engine overhead.
- **Transwarp**  October 2021 - March 2022
Database Development Intern Shanghai, China
 - Participated in the development of TimeLyre, a time series database based on a fork of InfluxDB.
 - Developed a benchmark tool for time-series data ingestion, data generation, and metric measurement.
 - Analyzed user-defined functions in InfluxDB and designed an interface for UDF authorization.
 - Implemented a storage-engine optimization that improved the compression ratio for time-series data.

SELECTED RESEARCH PROJECTS

- **IMBridge: [Impedance Mismatch Mitigation for Prediction Query Execution]** December 2023 - April 2024
Project Leader, in collaboration with [OceanBase](#) and Ant Group, accepted to SIGMOD/PACMMOD 2025 
 - Identified execution-mode mismatch between database engines and AI inference frameworks.
 - Proposed an inference-aware AI function operator with runtime context reuse to avoid repeated model setup.
 - Introduced batch-aware invocation to control inference batch sizes and improve accelerator utilization.
 - Built IMBridge on OceanBase and DuckDB, achieving over 10× speedup on application workloads.
- **IMLane: [Composable Framework for Efficient AI Function Execution in Database Engine]** May 2024 - June 2026
Project Leader, in collaboration with [OceanBase](#) and Ant Group, accepted to PVLDB 2026 Industry Track 
 - Identified parallelism and scheduling bottlenecks in database-native AI function execution.
 - Designed process-level parallel execution to avoid GIL contention and enable true parallel invocation.
 - Introduced decoupled scheduling to match AI functions with heterogeneous CPU/GPU resources.
 - Built a composable C++ framework integrated with OceanBase and DuckDB, achieving over 5× speedup over standard OceanBase execution.

- **RECS: [Scheduling Data Processing Pipelines for Incremental Recommendation Training]** *June 2023 - February 2025*
Core Project Member, in collaboration with [Tencent Inc.](#), accepted to SIGMOD 2025 Industrial Track
 - Identified resource idleness caused by intra-pipeline dependencies and inter-pipeline critical pipelines.
 - Proposed intra-pipeline scheduling with dynamic prefetching to overlap feature processing and embedding lookup.
 - Proposed critical-path-first inter-pipeline scheduling to reduce data processing time for incremental training.
 - Contributed to workload analysis, experiments, and paper revision; the prototype reduced training time by 20%.
- **ReTree: [Redundant Feature Test Elimination for Tree Model Inference]** *August 2025 - January 2026*
Project Member, accepted to SIGMOD 2026 Research Track
 - Characterized redundant feature tests in decision tree and random forest inference over SQL predicates.
 - Proposed predicate-aware elimination through subtree collapse and sliding subtree reorganization.
 - Integrated the optimization into DuckDB to reduce repeated comparisons in predicate filtering.
 - Contributed to problem formulation, analysis, and revision; the prototype achieved over 2× speedup.
- **Craftsman & EncoderForge: [Efficient ML-to-SQL Translation for ML Inference Pipelines]** *July 2024 - January 2026*
Project Member, in collaboration with [PingCAP](#), accepted to ICDE 2025 and SIGMOD 2026
 - Investigated inefficient SQL generation for encoders in machine learning inference pipelines.
 - Designed template-based fusion rules and composite-join translation for categorical encoders.
 - Proposed cost-based selection over CASE, JOIN, and hybrid SQL translation plans.
 - Contributed to problem definition, solution design discussions, baseline experiments, and paper revision; the prototypes achieved over 2× speedup.

ONGOING RESEARCH AND DEVELOPMENT

- **AI-driven High-Performance Database Engine Optimization** *May 2026 - Present*
Ongoing research direction
 - Exploring agent-driven database engine specialization, using workload understanding and query code synthesis to generate optimized execution paths and reduce general-purpose engine overhead.
- **SliceFlow: [Eliminating Redundant Computation for Recommendation Model Inference]** *September 2025 - Present*
Ongoing research project
 - Studying repeated input data patterns in recommendation ranking inference, where invariant user features are recomputed across candidate items.
 - Proposing ReDR, a repeated data representation that captures redundancy levels across tensors and operators.
 - Designing operator substitution and propagation-based graph transformation rules to separate invariant and variant computation.
 - Building a Torch.fx-based prototype to capture computation graphs, apply ReDR-guided rewrites, and evaluate inference performance gains.
- **Multimodal Data Processing Engine Benchmark** *April 2025 - Present*
Ongoing research exploration
 - Designing multimodal AI benchmark suites for Ray Data, Daft, Spark, Flink, and related data engines.
- **ZoteroOfMine: [Agent-assisted Research Workflow Toolkit]** *January 2026 - Present* 
Personal research tooling project
 - Developing a Zotero-based toolkit for paper reading, local CLI utilities, clipboard protocols, and agent-assisted literature workflows.

PUBLICATIONS

- [C.1] **Chenyang Zhang**, Junxiong Peng, Chen Xu*, Quanqing Xu, and Chuanhui Yang. (2024). IMBridge: Impedance Mismatch Mitigation between Database Engine and Prediction Query Execution. [In Companion of the 2024 International Conference on Management of Data \(SIGMOD\)](#).
- [C.2] **Chenyang Zhang**, Junxiong Peng, Chen Xu*, Quanqing Xu, and Chuanhui Yang. (2025). Mitigating the Impedance Mismatch between Prediction Query Execution and Database Engine. [In Proc. ACM Manag. Data \(PACMMOD\)](#).
- [C.3] Zihao Chen, **Chenyang Zhang**, Chen Xu*, Zhao Zhang, Jiaqiang Wang, Weining Qian, and Aoying Zhou. (2025). Scheduling Data Processing Pipelines for Incremental Training on MLP-based Recommendation Models. [In Companion of the 2025 International Conference on Management of Data \(SIGMOD\)](#).
- [C.4] Qingfeng Pan, Jiahe Zhi, **Chenyang Zhang**, Chen Xu*, Zhao Zhang, Anita Shao, Guanglei Bao, Qiu Cui, Xiaowei Chen, and Aoying Zhou. (2025). Machine Learning Inference Pipeline Execution Using Pure SQL Based on Operator Fusion. [In the 41st IEEE International Conference on Data Engineering \(ICDE\)](#).

[C.5] Mingxi Liu, Zhengyuan Ding, **Chenyang Zhang**, Qingfeng Pan, Huayou Su, Zhao Zhang, Chen Xu*, and Qingsong Ruan. (2026). Eliminating Redundant Feature Tests in Decision Tree and Random Forest Inference on SQL Predicates. In *Proc. ACM Manag. Data (PACMMOD)*.

[C.6] Qingfeng Pan, Qingyuan Jing, **Chenyang Zhang**, Jiahe Zhi, Chen Xu*, and Heng Long. (2026). EncoderForge: Generating Efficient SQL for Encoders in Machine Learning Inference Pipelines. In *Proc. ACM Manag. Data (PACMMOD)*.

[C.7] **Chenyang Zhang**, Linjun Lu, Qingfeng Pan, Chen Xu*, Xianzhong Cao, Quanqing Xu, and Chuanhui Yang. (2026). IMLane: Composable Framework for Efficient AI Function Execution in Database Engine. Accepted to PVLDB 2026 Industry Track.

RESEARCH AND TECHNICAL SKILLS

- **Programming Languages:** SQL, Python, C++, Go, Java, Scala, JavaScript/TypeScript.
- **Systems and Frameworks:** DuckDB, OceanBase, InfluxDB, TensorFlow, PyTorch, Torch.fx, ONNX Runtime, Ray, vLLM, SGLang.
- **Research Skills:** Literature review, problem formulation, system prototyping, experiment design, workload analysis, performance profiling, evaluation methodology, result interpretation, and academic writing.
- **AI and Agent Tooling:** AI coding, LLM agents, tool calling, MCP, RAG, workflow orchestration.

HONORS AND AWARDS

- | | |
|--|---------------|
| • ECNU Academic Innovation Promotion Program for Excellent Doctoral Students
<i>East China Normal University</i> | July 2026 |
| • SIGMOD 2025 Student Travel Grant
<i>SIGMOD/PODS 2025 Organizing Committee</i> | June 2025 |
| • SIGMOD 2024 Student Support Scholarship
<i>SIGMOD/PODS 2024 Organizing Committee</i> | June 2024 |
| • Outstanding Graduate of Shanghai
<i>Shanghai Municipal Commission of Education</i> | July 2022 |
| • Outstanding Undergraduate Thesis
<i>Donghua University</i> | June 2022 |
| • First Prize, AI Application Track
<i>China Undergraduate Computer Design Competition</i> | August 2021 |
| • Donghua University Student Scholarship
<i>Donghua University Education Foundation</i> | November 2020 |
| • Outstanding Student
<i>Donghua University</i> | November 2020 |

VOLUNTEER EXPERIENCE

- | | |
|---|--------------------------------|
| • Teaching Assistant
<i>Graduate Course: Distributed Computation Systems</i> <ul style="list-style-type: none">◦ Participated in discussions and evaluations of student projects.◦ Explored recent ideas in data processing systems through project discussions. | September 2024 - December 2024 |
|---|--------------------------------|